AN ETALE MORPHISM ON TOP OF A FORMALLY UNRAMIFIED MORPHISM STAYS FORMALLY UNRAMIFIED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2566-2602 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2604-2654, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally unramified morphisms and their lifting criterion, and the lifting property of etale morphisms; the argument is retained.

Source reference. Original lemma etale-theorem-remarkable-equivalence: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2654–2654. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-composition-formally-smooth-etale-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2486–2495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-representable-property-formally-property: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2529–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $F, G, H : (Sch/S)^{opp}_{fppf} \to Sets$. Let $a : F \to G$, $b : G \to H$ be transformations of functors. Assume that a is representable, surjective, and étale.*

(1) *If b is formally smooth, then $b \circ a$ is formally smooth.*
(2) *If b is formally étale, then $b \circ a$ is formally étale.*
(3) *If b is formally unramified, then $b \circ a$ is formally unramified.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Conversely, consider a solid commutative diagram

$$\begin{array}{ccc} G & \longleftarrow & T \\ {\scriptstyle b}\downarrow & \nearrow & \downarrow{\scriptstyle i} \\ H & \longleftarrow & T' \end{array}$$

with T' an affine scheme over S and $i : T \to T'$ a closed immersion defined by an ideal of square zero.

(4) *If $b \circ a$ is formally smooth, then for every $t \in T$ there exists an étale morphism of affines $U' \to T'$ and a morphism $U' \to G$ such that*

$$\begin{array}{ccccc} G & \longleftarrow & T & \longleftarrow & T \times_{T'} U' \\ {\scriptstyle b}\downarrow & \swarrow & & & \downarrow \\ H & \longleftarrow & T' & \longleftarrow & U' \end{array}$$

commutes and t is in the image of $U' \to T'$.

(5) *If $b \circ a$ is formally unramified, then there exists at most one dotted arrow in the diagram above, i.e., b is formally unramified.*

(6) *If $b \circ a$ is formally étale, then there exists exactly one dotted arrow in the diagram above, i.e., b is formally étale.*

Proof. Assume b is formally smooth (resp. formally étale, resp. formally unramified). Since an étale morphism is both smooth and unramified we see that a is representable and smooth (resp. étale, resp. unramified). Hence parts (1), (2) and (3) follow from a combination of Lemma lemma-representable-property-formally-property and Lemma lemma-composition-formally-smooth-etale-unramified.

Assume that $b \circ a$ is formally smooth. Consider a diagram as in the statement of the lemma. Let $W = F \times_G T$. By assumption W is a scheme surjective étale over T. By Étale Morphisms, Theorem etale-theorem-remarkable-equivalence there exists a scheme W' étale over T' such that $W = T \times_{T'} W'$. Choose an affine open subscheme $U' \subset W'$ such that t is in the image of $U' \to T'$. Because $b \circ a$ is formally smooth we see that the exist morphisms $U' \to F$ such that

$$\begin{array}{ccccc} F & \longleftarrow & W & \longleftarrow & T \times_{T'} U' \\ {\scriptstyle b\circ a}\downarrow & \swarrow & & & \downarrow \\ H & \longleftarrow & T' & \longleftarrow & U' \end{array}$$

commutes. Taking the composition $U' \to F \to G$ gives a map as in part (5) of the lemma.

Assume that $f, g : T' \to G$ are two dotted arrows fitting into the diagram of the lemma. Let $W = F \times_G T$. By assumption W is a scheme surjective étale over T. By Étale Morphisms, Theorem etale-theorem-remarkable-equivalence there exists a scheme W' étale over T' such that $W = T \times_{T'} W'$. Since a is formally étale the compositions

$$W' \to T' \xrightarrow{f} G \quad \text{and} \quad W' \to T' \xrightarrow{g} G$$

lift to morphisms $f', g' : W' \to F$ (lift on affine opens and glue by uniqueness). Now if $b \circ a : F \to H$ is formally unramified, then $f' = g'$ and hence $f = g$ as $W' \to T'$ is an étale covering. This proves part (6) of the lemma.

Assume that $b \circ a$ is formally étale. Then by part (4) we can étale locally on T' find a dotted arrow fitting into the diagram and by part (5) this dotted arrow is unique. Hence we may glue the local solutions to get assertion (6). Some details omitted. □